

GATE SOLUTION IN MINING ENGINEERING

(Containing Solved Questions 2015 - 2007)

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Gate Solution-2010

- Q. 1 Ascensionally ventilated coal mine inclines ideally should have higher methane layering number when compared to descensionally ventilated inclines. The reason is
- (A) in ascensionally ventilated inclines density of air is higher
 - (B) ascensionally ventilated inclines creates conditions for improved turbulent mixing of methane layer
 - (C) methane drainage is not practiced in ascensionally ventilated inclines
 - (D) descensionally ventilated inclines creates conditions for improved turbulent mixing of methane layer

Answer (D)

- Q.2 A coolant is a desirable component in the design of a Self Contained Breathing apparatus since
- (A) surrounding can be hot and humid during rescue
 - (B) a rescue worker generates large amount of metabolic heat
 - (C) exhaled air CO_2 absorption is an exothermic reaction
 - (D) exhaled air water vapour has to be condensed

Answer (C)

- Q.3 determine the correctness or otherwise of the following Assertion [a] and the Reason [r]

Assertion : Both intake and return side stoppings must be closed simultaneously in the event of sealing off a coal mine panel with explosion hazard following a fire.

Reason : By continuously ventilating the area till simultaneous closure of the stopping, the possibility of an explosion hazard due to gas build up is avoided.

- (A) [a] is true but [r] is false
 (B) Both [a] and [r] are true and [r] is the correct reason for [a]
 (C) Both [a] and [r] are true and [r] is not the correct reason for [a]
 (D) Both [a] and [r] are false

Answer (C)

- Q.4 In a Cartesian coordinate system the vertices of a triangular plate are given by $(-2, 1)$, $(3, 4)$ and $(-4, -8)$. The coordinates of the centre of gravity of the plate are
- (A) 3, 4 (B) 7, 12 (C) $-1, -1$ (D) $-3, -4$

Answer (C)

Solution .

The coordinates of center of gravity (O) of triangular plate ABC, having three vertices (x_1, y_1) , (x_2, y_2) , (x_3, y_3) are given by

$$O_x = \frac{x_1 + x_2 + x_3}{3}, O_y = \frac{y_1 + y_2 + y_3}{3}$$

$$\therefore O_x = \frac{-2 + 3 - 4}{3}, O_y = \frac{1 + 4 - 8}{3}$$

$$\therefore (-1, -1)$$

- Q.5 An air quality parameter required to monitored under the Indian National Ambient Air Quality Standards is
- (A) As (B) Pb (C) Hg (D) Silica

Answer (B)

- Q.6 In an underground coal mine, a freshly exposed roof can be supported by a temporary support in the form of
- (A) triangular chocks
 (B) crew props
 (C) safari support

(D) hydraulic prop

Answer (C)

Q.7 At a surface mine office the independent sound pressure level (SPL) measured in dB (A) on account of 3 drill machines are 85, 88 and 85. If all the three machines work simultaneously, the combined SPL, in dB (A), is

(A) 91 (B) 90 (C) 92 (D) 94

Answer (A)

Solution .

Combined SPL is given by :-

$$L_{com} = 10 \log_{10} \left[\sum_{i=1}^n t_i 10^{\frac{L_i}{10}} \right]$$

$$L_{com} = 10 \log_{10} \left[1 \times 10^{\frac{85}{10}} + 1 \times 10^{\frac{88}{10}} + 1 \times 10^{\frac{85}{10}} \right]$$

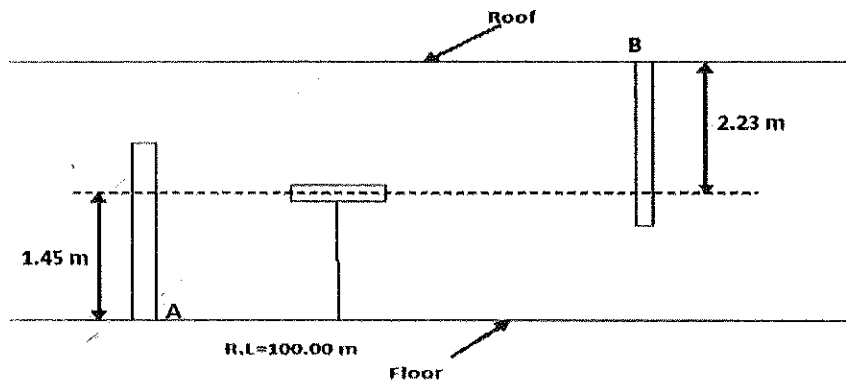
$$L_{com} = 91 \text{ dB}$$

Q.8 The backsight reading on a bench mark of RL 100.0 m is 1.45 m. if the inverse staff reading on a foresight is 2.23 m, the RL of the staff station in m is

(A) 105.13 (B) 103.68 (C) 100.78 (D) 98.55

Answer (B)

Solution .



$$\begin{aligned}\therefore \text{R.L. of staff station B is} &= 100 + 1.45 + 2.23 \\ &= 103.68 \text{ m}\end{aligned}$$

Q.9 For a mine of production t per year, the total cost of production is given by $at^2 + b$. The revenue from sale is given by ct . if a , b and c , are constant, the breakeven value of t is

- (A) $\left[c \pm \sqrt{c^2 - 4ab} \right] / (2a)$ (B) $\left[\sqrt{c^2 - 4ab} \right] / (2a)$
 (C) $\left[-c \pm \sqrt{c^2 - 4ab} \right] / (2a)$ (D) $\left[c \pm \sqrt{c^2 + 4ab} \right] / (2a)$

Answer (A)

Solution .

In case of breakeven condition -

Total cost of production = Revenue

$$at^2 + b = ct$$

$$\text{Or } at^2 - ct + b = 0$$

$$\therefore t = \frac{c \pm \sqrt{c^2 - 4ab}}{2a}$$

Q.10 The value of the limit $\lim_{x \rightarrow 1} \left[\frac{1 - x^{-1/3}}{1 - x^{-2/3}} \right]$ is

- (A) ∞ (B) 1 (C) 0 (D) $\frac{1}{2}$

Answer (D)

Solution .

$$f(x) = \lim_{x \rightarrow 1} \left[\frac{1 - x^{-1/3}}{1 - x^{-2/3}} \right]$$

$$f(x) = \lim_{x \rightarrow 1} \left[\frac{\frac{x^{1/3} - 1}{x^{1/3}}}{\frac{x^{2/3} - 1}{x^{2/3}}} \right]$$

$$f(x) = \lim_{x \rightarrow 1} \frac{x^{1/3}(x^{1/3} - 1)}{x^{2/3} - 1}$$

$$f(x) = (1)^{1/3} \lim_{x \rightarrow 1} \left\{ \frac{x^{1/3} - 1}{x - 1} \cdot \frac{x - 1}{x^{2/3} - 1} \right\}$$

$$f(x) = \frac{\lim_{x \rightarrow 1} \frac{x^{1/3} - 1}{x - 1}}{\lim_{x \rightarrow 1} \frac{x^{2/3} - 1}{x - 1}}$$

$$\left[\text{since, } \lim_{x \rightarrow a} \frac{x^n - a^n}{x - a} = n a^{n-1} \right]$$

$$f(x) = \frac{\frac{1}{3} \cdot (1)^{\frac{1}{3}-1}}{\frac{2}{3} \cdot (1)^{\frac{2}{3}-1}}$$

$$\therefore f(x) = 1/2$$

Q.11 Two determinants of order n are multiplied. The order of the resultant determinant is

- (A) n (B) 2n (C) n^2 (D) $n/2$

Answer (A)

Solution .

Note - The product of two determinant of n^{th} order is a determinant of the n^{th} order.

Therefore, Two determinant of order n are multiplied then the order of resultant determinant is n.

Q.12 The partial differential equation, $r \frac{\partial \theta}{\partial r} = \text{constant}$, is a solution for

- (A) $\frac{\partial^2 \theta}{\partial r^2} - \frac{1}{r} \frac{\partial \theta}{\partial r} = 0$ (B) $\frac{\partial^2 \theta}{\partial r^2} + \frac{\partial \theta}{\partial r} = 0$

$$(C) \quad r^2 \frac{\partial^2 \theta}{\partial r^2} + \frac{1}{r} \frac{\partial \theta}{\partial r} = 0$$

$$(D) \quad \frac{\partial^2 \theta}{\partial r^2} + \frac{1}{r} \frac{\partial \theta}{\partial r} = 0$$

Answer (D)

Solution .

We have,

$$r \frac{\partial \theta}{\partial r} = \text{constant}$$

Differentiating w.r.t, r , then

$$r \frac{\partial^2 \theta}{\partial r^2} + \frac{\partial \theta}{\partial r} = 0$$

Dividing both sides by $1/r$, then

$$\frac{r \partial^2 \theta}{r \partial r^2} + \frac{1}{r} \frac{\partial \theta}{\partial r} = 0$$

$$\frac{\partial^2 \theta}{\partial r^2} + \frac{1}{r} \frac{\partial \theta}{\partial r} = 0$$

Q.13 In mohr-coulomb failure criteria , the ratio of the uniaxial compressive strength to the tensile strength is

$$(A) \quad \frac{1 + \sin \phi}{1 - \sin \phi}$$

$$(B) \quad \frac{1 - \sin \phi}{1 + \sin \phi}$$

$$(C) \quad \frac{C (1 + \sin \phi)}{(1 - \sin \phi)}$$

$$(D) \quad \frac{2C (1 + \sin \phi)}{(1 - \sin \phi)}$$

Answer (A)

Solution .

Uniaxial compressive strength is given by

$$\sigma_c = \frac{2c \cos \phi}{1 - \sin \phi} \dots\dots\dots(1)$$

And

Tensile strength is given by

$$\sigma_t = \frac{2c \cos \phi}{1 + \sin \phi} \dots\dots\dots(2)$$

From equa. (1) & (2),

$$\frac{\sigma_c}{\sigma_t} = \frac{1 + \sin \phi}{1 - \sin \phi}$$

Q.14 The average young's modulus and poisson's ratio values of a limestone sample are 60×10^3 MPa and 0.3 respectively. The shear modulus in MPa is

- (A) 23.07 (B) 230.7 (C) 2307.0 (D) 23070.0

Answer (D)

Solution .

$$\text{Poisson ratio} = \frac{1}{m} = 0.3$$

$$\text{Or} \quad m = 3.33$$

$$\text{And} \quad \text{young modulus} = E = 60 \times 10^3 \text{ MPa}$$

We have,

$$\text{Shear modulus} = C = \frac{m E}{2(m + 1)}$$

$$C = \frac{3.33 \times 60 \times 10^3}{2 \times (3.33 + 1)}$$

$$C = 23071 \text{ MPa}$$

Q.15 The angle of draw in a trough subsidence helps in determining the

- (A) Maximum subsidence
(B) Extent of surface subsidence
(C) Plane of failure
(D) Critical width of opening

Answer (B)

Q.16 Recapping a winding rope is done due to

- (A) increase the strength flexural strength of rope
- (B) increase the flexibility of the rope
- (C) remove a portion of rope subjected to deterioration
- (D) prevent the rope from excessive rusting

Answer (C)

Q.17 Match the following for standard diamond drill rods.

Specification	Outer diameter in mm
P. AW	p. 34.9
Q. BW	q. 44.4
R. EW	r. 54.0
S. NW	s. 66.7
(A) P-r, Q-q, R-s, S-p	(B) P-r, Q-p, R-s, S-q
(C) P-q, Q-r, R-p, S-s	(D) P-q, Q-r, R-s, S-p

Answer (C)

Q.18 Payback period is time required

- (A) for the cash income from a project to get back the initial cash investment
- (B) from the start of the project to the time to recover the total initial investment
- (C) from the start of the project to the start of the production
- (D) to the period during which internal rate of return is generated

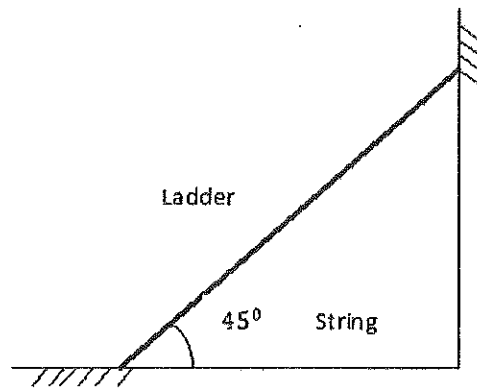
Answer (B)

Q.19 For electric signaling systems in underground coal mines, the statement that is NOT true is

- (A) all signaling equipment must be intrinsically safe
- (B) the signaling circuit must be connected to ground
- (C) the source of current should be an approved dry battery
- (D) DC bells or relay when connected in parallel should be supplied from a single source current

Answer (D)

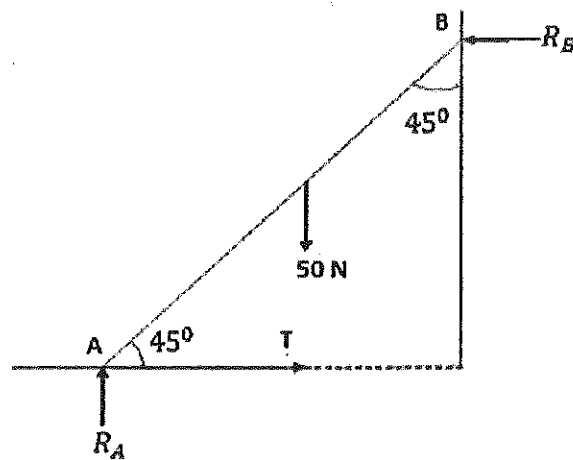
- Q.20 A ladder of weight 50 N rests against a frictionless wall and floor as shown in figure. A horizontal string ties the base of the ladder to the wall. The tension in the string in N is



- (A) 100 (B) 50 (C) 72 (D) 25

Answer (D)

Solution .



$$R_A = R_B = \frac{W}{2} = \frac{50}{2} = 25 \text{ N}$$

$$\text{Now, } \sum H = 0$$

$$T - R_B = 0 \quad [\text{where } T \text{ is the tension in the string}]$$

$$\therefore T = R_B = 25 \text{ N}$$

- Q.21** The mean and standard deviation of the grade of ore in a deposit are 62% and 5% respectively. The coefficient of variation of the grade in % is
- (A) 24.8 (B) 12.4 (C) 8.0 (D) 4.0

Answer (C)

Solution .

We have,

$$\begin{aligned}\text{Coefficient of variation} &= \frac{\text{standard deviation}}{\text{mean}} \times 100 \\ &= \frac{5}{62} \times 100 \\ &= 8 \%\end{aligned}$$

- Q.22** The variance of failure time (time to failure) of an electric motor in shovel is 1600 hr^2 . If the failure time follows an exponential distribution, the expected failure time in hr is
- (A) 40 (B) 80 (C) 800 (D) 1600

Answer (A)

Solution .

The expected value of exponential distribution is given by $\frac{1}{\mu}$ and its variance is given by $\frac{1}{\mu^2}$

$$\therefore \text{variance} = 1600 \text{ hr}^2$$

$$\therefore \left(\frac{1}{\mu}\right)^2 = 1600 \text{ hr}^2$$

$$\text{Or } \frac{1}{\mu} = \sqrt{1600 \text{ hr}}$$

$$\therefore \frac{1}{\mu} = 40 \text{ hr}$$

- Q.23** Match the following

Instrument

Purpose/M Measurement

1. Abneys level

a. horizontal and vertical angles

2. Pantograph
3. Planimeter
4. Box sextant

- b. area of plotted figures
- c. enlargement and reduction of plotted maps
- d. angle of inclination

- (A) 1-a, 2-c, 3-d, 4-b
 (B) 1-c, 2-b, 3-d, 4-a
 (C) 1-b, 2-a, 3-d, 4-c
 (D) 1-d, 2-c, 3-b, 4-a

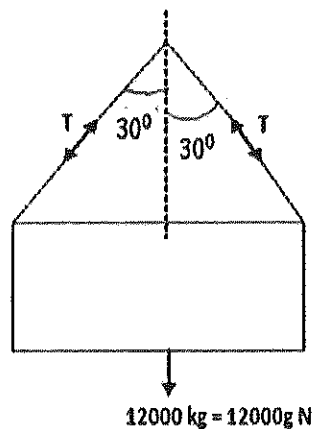
Answer (D)

Q.24 A cage of weighing 12000 kg is raised by four chains each making an angle of 30° with the vertical. The tension in each chain in KN is

- (A) 41 (B) 34 (C) 25 (D) 20

Answer (B)

Solution .



Since cage is raised by four chains and each making an angle of 30° with vertical. So vertical component of each chain will be $T \cos 30^\circ$.

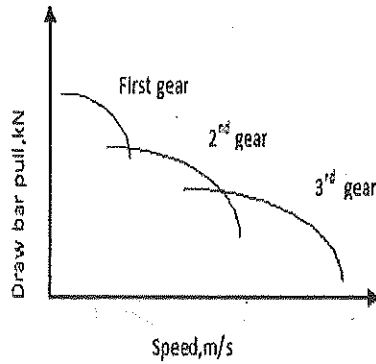
$$\therefore \sum V = 0$$

$$T \cos 30^\circ + T \cos 30^\circ + T \cos 30^\circ + T \cos 30^\circ - 12000 \times 9.81 = 0$$

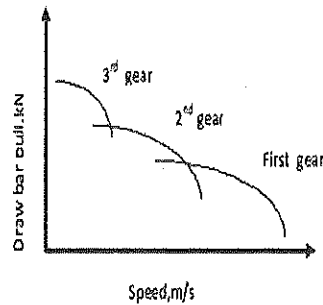
$$\therefore 4T \cos 30^\circ = 12000 \times 9.81$$

$$T = 33982 \text{ N} = 34 \text{ kN}$$

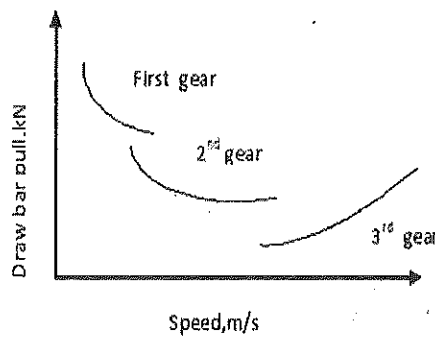
Q.25 The relationship between the drawbar pull and speed for different gears of a self propelling vehicle is represented by



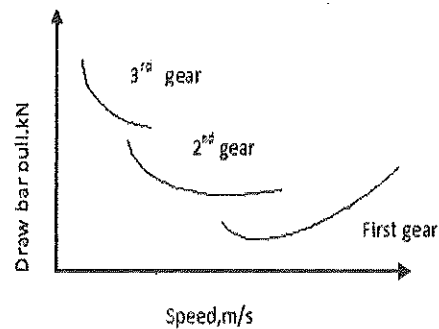
(P)



(Q)



(R)



(S)

(A) Q

(B) S

(C) R

(D) P

Answer (D)

Q.26 A flammable mixture has 70% CH_4 and 30% CO. The lower flammability limit for these gases are 5% and 13% respectively. For the mixture, the lower flammability limit in % is

(A) 6.13

(B) 8.72

(C) 10.25

(D) 12.16

Answer (A)

Solution .

The lower flammability limit in (%) for mixture is given by:-

$$X = \left[\frac{100}{\frac{P_1}{N_1} + \frac{P_2}{N_2} + \frac{P_3}{N_3} + \dots + \frac{P_n}{N_n}} \right] \text{percent}$$

$$X = \left[\frac{100}{\frac{70}{5} + \frac{30}{13}} \right] \%$$

$$X = 6.13 \%$$

Q.27 The volume of tetrahedron with vertices at (0,0,0),(1,0,0),(0,1,0) and (0,0,1) is

(A) 1/2

(B) 1/4

(C) 1/6

(D) 1/8

Answer (C)

Solution .

$$\text{Volume of tetrahedron} = \frac{1}{3} \times A_0 \times h$$

Where, A_0 - area of base

h - height from base to apex

Assuming apex vertice, (A) = (0, 0, 0)

and base vertices ,B = (1, 0, 0)

$$C = (0, 1, 0)$$

$$D = (0, 0, 1)$$

Now,

$$a = BC = \sqrt{(1-0)^2 + (0-1)^2 + (0-0)^2} = \sqrt{2}$$

$$b = CD = \sqrt{(0-0)^2 + (1-0)^2 + (0-1)^2} = \sqrt{2}$$

$$c = DB = \sqrt{(0-1)^2 + (0-0)^2 + (1-0)^2} = \sqrt{2}$$

$$s = \frac{a+b+c}{3} = \frac{\sqrt{2}+\sqrt{2}+\sqrt{2}}{3} = 2.121$$

$$\text{Area of base} = \sqrt{s(s-a)(s-b)(s-c)}$$

$$\therefore \text{Area of base} = \sqrt{2.121(2.121 - \sqrt{2})(2.121 - \sqrt{2})(2.121 - \sqrt{2})} = 0.8653$$

$$\begin{aligned} \text{Now coordinates of the centre of base are} &= \frac{1+0+0}{3}, \frac{0+1+0}{3}, \frac{0+0+1}{3} \\ &= \frac{1}{3}, \frac{1}{3}, \frac{1}{3} \end{aligned}$$

$$\text{Now, height from base to apex (h)} = \sqrt{\left(0 - \frac{1}{3}\right)^2 + \left(0 - \frac{1}{3}\right)^2 + \left(0 - \frac{1}{3}\right)^2}$$

$$h = 0.5773$$

$$\text{Hence, area of tetrahedron} = \frac{1}{3} \times 0.8653 \times 0.5773 = \frac{1}{6}$$

Q.28 A balanced winder raises 3000 tonnes per day from a depth of 500 m. The payload of the winding cage is 7 tonnes. The energy consumed per day in kWh at 70% winder efficiency is

(A) 6030

(B) 5840

(C) 5750

(D) 5630

Answer (B)

Solution .

Material hoisting = 3000 tonnes per day

$$= \frac{3000 \times 1000 \times 9.81}{24 \times 60 \times 60} \frac{N}{s}$$

$$= \frac{3000 \times 1000 \times 9.81 \times 500}{24 \times 60 \times 60} \frac{Nm}{s}$$

$$= 170312.5 \text{ watt}$$

$$= 170.3125 \text{ kW}$$

$$= \frac{170.3125}{h} \text{ kWh}$$

$$= \frac{170.3125 \times 24}{\text{day}} \text{ kWh}$$

$$= 4087.5 \frac{\text{kWh}}{\text{day}}$$

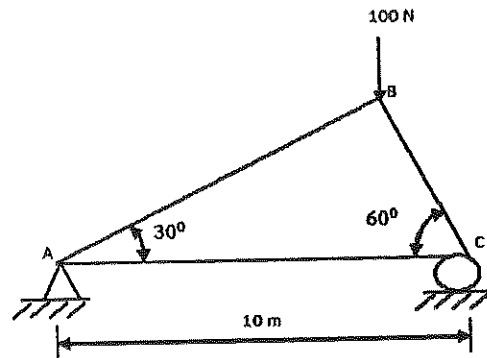
But winder efficiency is 70 %

$$= \frac{4087.5 \text{ kwh}}{0.70 \text{ day}}$$

$$= 5839.2857 \text{ kwh/day}$$

$$= 5840 \text{ kwh /day}$$

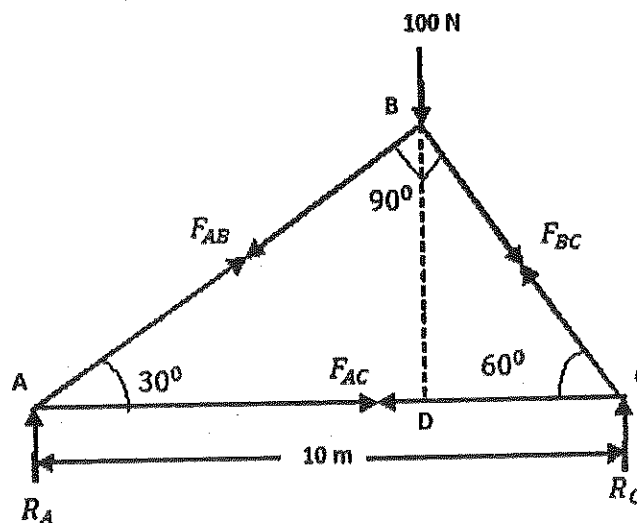
Q.29 A truss is loaded as shown in the figure. The force in the member AC is



- (A) tension 75.9 N
- (B) compression 43.3 N
- (C) tension 43.4 N
- (D) compression 75.9 N

Answer (C)

solution .



Applying sine formula

$$\frac{10}{\sin 90^\circ} = \frac{AB}{\sin 60^\circ} = \frac{BC}{\sin 30^\circ}$$

$$\therefore AB = 8.66 \text{ m} \text{ \& } BC = 5 \text{ m}$$

$$\text{Now, } \cos 60^\circ = \frac{CD}{5}$$

$$\therefore CD = 2.5 \text{ m}$$

$$\text{And } AD = 10 - 2.5 = 7.5 \text{ m}$$

$$\text{Now, } \sum M_A = 0$$

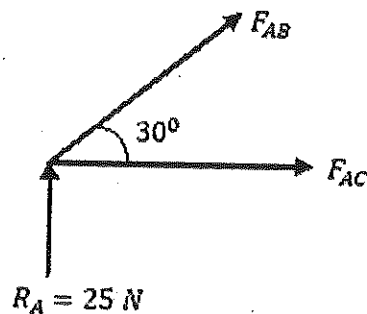
$$-100 \times 7.5 + R_c \times 10 = 0$$

$$R_c = 75 \text{ N}$$

$$\text{And } \sum V = 0$$

$$R_A + R_c - 100 = 0$$

$$R_A = 25 \text{ N}$$



$$\sum V = 0$$

$$25 + F_{AB} \times \sin 30^\circ = 0$$

$$\therefore F_{AB} = -50 \text{ N}$$

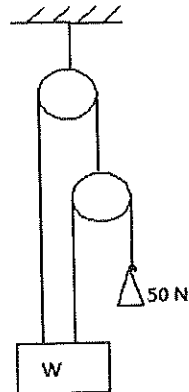
$$\sum H = 0$$

$$F_{AC} + F_{AB} \cos 30^\circ = 0$$

$$F_{AC} = -F_{AB} \cos 30^\circ$$

$$\therefore F_{AC} = 43.3 \text{ N (Tension)}$$

Q.30 In the frictionless pulley system shown in the figure, each pulley weighs 20 N. The weight W in N, that can be lifted by the system under the conditions shown is



(A) 200

(B) 170

(C) 150

(D) 100

Answer (B)

Solution .

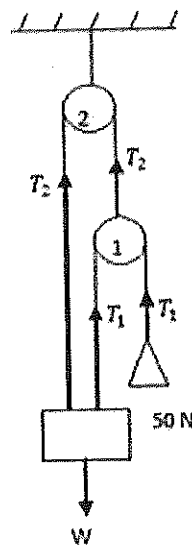


Fig-01

Here, $T_1 = 50\text{ N}$

From figure 01.

Equilibrium of load W

$$\sum V = 0$$

$$T_1 + T_2 - W = 0$$

$$W = T_1 + T_2 \dots \dots \dots (1)$$

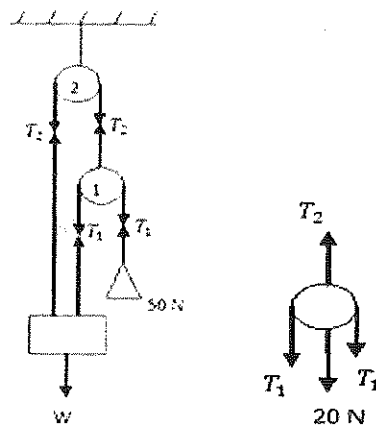


Fig.-02

From figure (2), equilibrium of pulley 1

$$\sum V = 0$$

$$T_2 - T_1 - T_1 - 20 = 0$$

$$T_2 = 2T_1 + 20$$

$$T_2 = 2 \times 50 + 20$$

$$T_2 = 120\text{ N}$$

From equation (1)

$$W = 50 + 120$$

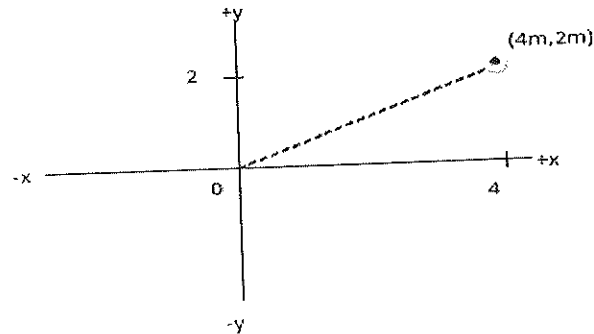
$$W = 170\text{ N}$$

Q.31 A force of $50\hat{i} - 50\hat{j}$ N is moved from the origin to a coordinate (4.0m, 2.0m). The work done in the process in J is

- (A) 75.6 (B) 85.5 (C) 90.2 (D) 100.0

Answer (D)

solution .



$$\text{Displacement} = \vec{d} = 4\hat{i} - 2\hat{j}, \text{m}$$

$$\text{Force} = \vec{F} = 50\hat{i} - 50\hat{j}, \text{N}$$

$$\begin{aligned} \therefore \text{Work done} &= \vec{F} \cdot \vec{d} \\ &= (50\hat{i} - 50\hat{j}) \cdot (4\hat{i} - 2\hat{j}) \\ &= 200 - 100 \\ &= 100 \text{ J} \end{aligned}$$

Q.32 The queue of trucks at a crusher plant hopper is known to be M/M/1 queue. The probability that there is no truck to unload is 0.3. Due to rains the mean service time at the hopper is increased by 30%. As a consequence, the expected number of trucks in the queuing system (including the one possibly unloading) become

- (A) 10 (B) 12 (C) 14 (D) 16

Answer (A)

Solution .

$$P_0 = (1 - \rho) = 1 - \frac{\lambda}{\mu}$$

$$0.3 = 1 - \frac{\lambda}{\mu}$$

$$\therefore \lambda = 0.7 \mu = \text{mean arrival rate (per unit time)}$$

Since mean service time $\left(\frac{1}{\mu}\right)$ is increased by 30 %,

$$\text{So, new mean service time} = \frac{1}{\mu'} = \frac{1}{\mu} \times \frac{(100 + 30)}{100} = \frac{1.3}{\mu}$$

$$\text{Mean service rate} = \mu' = \frac{\mu}{1.3} \text{ per unit time}$$

We have,

$$\text{Number of trucks in the queue system} = N_Q = \frac{\lambda}{\mu' - \lambda}$$

$$N_Q = \frac{0.7 \mu}{\frac{\mu}{1.3} - 0.7 \mu}$$

$$\therefore N_Q = 10$$

Q.33 Pull from a underground tunnel blasting is normally distributed with a mean of 100 tonnes and variance $100 (\text{tonnes})^2$. The probability that the tonnage value from a blast exceed 110 is

(A) 0.60

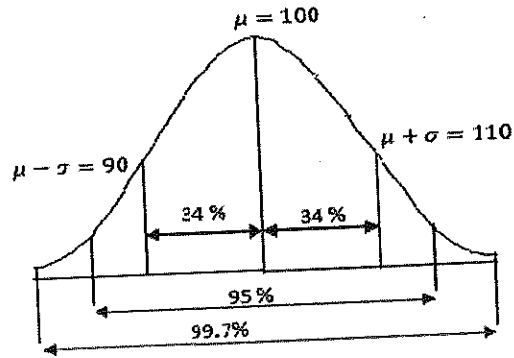
(B) 0.80

(C) 0.16

(D) 0.32

Answer (C)

solution .



Mean = $\mu = 100$

Variance = $100(\text{tonne})^2$

$\therefore$ standard deviation = $\sigma = 10 \text{ tonne}$

Hence, the probability that the tonnage value from a blast exceeds 110 is :-

$$(100 - 50 - 34) = 16\% = 0.16$$

Q.34 The feasibility region of an LP problem in variable x and y is given by the following constraints, in addition to the non-negativity constraints.

$$y \leq 60; x \leq 90; x + y \leq 70.$$

The number of corner point feasible solutions for this problem are

(A) 3

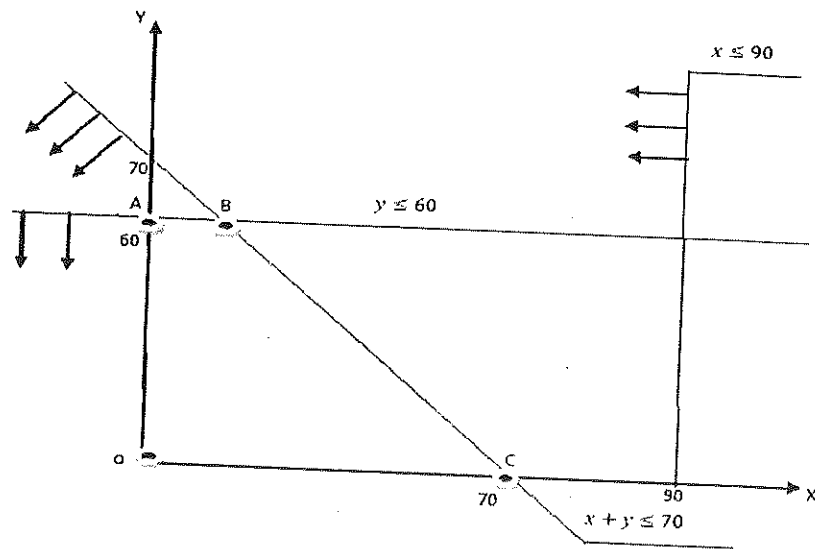
(B) 4

(C) 5

(D) 6

Answer (B)

solution .



Hence, corner point feasible solution will be 4 i.e. O,A,B,C.

Q .35 The unit cost matrix of a balanced transportation problem is shown below

Source	Destination			supply
	D_1	D_2	D_3	
S_1	7	3	6	60
S_2	5	4	9	60
S_3	8	6	7	80
Demand	50	120	30	

The transportation cost of the initial basic feasible solution obtained by the North – West rule is

- (A) 1025 (B) 1075 (C) 1130 (D) 1226

Answer (C)

Solution .

Destination		D_1	D_2	D_3	Supply
Source	S_1	7 (50)	3 (10)	6	60/10/0
	S_2	5	4 (60)	9	60/0
	S_3	8	6 (50)	7 (30)	80/30/10
Demand		50/0	120/110/50	30/0	

$$\begin{aligned}\text{Hence, transportation cost} &= 7 \times 50 + 3 \times 10 + 4 \times 60 + 6 \times 50 + 7 \times 30 \\ &= 1130\end{aligned}$$

- Q. 36 A high volume air sampler is operated for 8 hours in a mine with the flow rate of air varying from $1.5 \text{ m}^3/\text{min}$ to $1.3 \text{ m}^3/\text{min}$. The empty weight of the filter paper is 2.30 g and the final weight is 2.65 g. The mean concentration of the suspended particulate matter (SPM) during the study period in $\mu\text{g}/\text{m}^3$ is
- (A) 591 (B) 550 (C) 545 (D) 521

Answer (D)

Solution .

Total suspended solids/suspended particulate matter is given by-

$$\text{TSS/SPM} = \frac{(A - B)}{C}, \mu\text{g}/\text{m}^3$$

Where, A- weight of filter plus retained solids in $\mu\text{g} = 2.65 \text{ gm} = 2.65 \times 10^6 \mu\text{g}$

B- weight of clean filter in $\mu\text{g} = 2.30 \text{ gm} = 2.30 \times 10^6 \mu\text{g}$

C- volume of sample filtered in m^3

$$\text{Now, } C = \frac{1.5 + 1.3}{2} = 1.4 \text{ m}^3/\text{minute}$$

$$\therefore 1 \text{ minute} = 1.4 \text{ m}^3$$

$$\therefore 8 \text{ hour} = 8 \times 1.4 \times 60 \text{ m}^3$$

Therefore,

$$\text{TSS/SPM} = \frac{2.65 - 2.30}{1.4 \times 8 \times 60} \times 10^6 = 521 \mu\text{g}$$

Q .37 A developed panel for a coal seam having incubation period of 6 months has 32 square pillars under extraction having size 25 m, and height 3.0 m. Density of the coal is 1.4 tonne/m³. Extraction ratio during depillaring is expected to be 75%. To depillar the panel within the incubation period, assuming 25 workings days in a month, the production from the panel in tone/day is

(A) 420

(B) 480

(C) 560

(D) 680

Answer (A)

Solution .

Production from one pillar = $25 \times 25 \times 3 \times 1.4 = 2625$ tonnes

So, production from 32 pillars = $32 \times 2625 = 84000$ tonnes

$\therefore$ Production of panel within incubation period ($6 \times 25 = 150$ days) = 84000 tonnes

Therefore ,

Production from panel with 75% percentage of extraction in tonne/day is:-

$$= \frac{84000}{150} \times 0.75 = 420 \text{ tonnes /day}$$

Q.38 A closed traverse ABCDE of perimeter 425 m has a total error +0.25 m in latitude and -0.44m in departure. The precision of traverse is

(A) 1 in 556

(B) 1 in 775

(C) 1 in 833

(D) 1 in 1024

Answer (C)

Solution .

We know,

$$\text{Closing error} = \sqrt{(\text{latitude})^2 + (\text{longitude})^2}$$

$$\text{Closing error} = \sqrt{(0.25)^2 + (-0.44)^2}$$

$$\text{Closing error} = 0.51 \text{ m}$$

Since 425 m has a error of 0.51 m

$$1 \text{ m has a closing error of } \frac{0.51}{425} \text{ or } \frac{1}{425/0.51} \text{ or } \frac{1}{833}$$

Hence , precision of traverse is 1 in 833.

Q.39 The value of the given integral is

$$\int_{\frac{\pi}{5}}^{\frac{3\pi}{10}} \frac{\sin x}{(\sin x + \cos x)} dx$$

(A) $\frac{\sin \pi/8}{10}$

(B) $\frac{\pi}{10}$

(C) $\frac{\sin \pi/5}{10}$

(D) $\frac{3\pi}{10}$

Solution .

$$\text{Let } I = \int_{\frac{\pi}{5}}^{\frac{3\pi}{10}} \frac{\sin x}{\sin x + \cos x} dx \dots\dots\dots (1)$$

$$I = \int_{\frac{\pi}{5}}^{\frac{3\pi}{10}} \frac{\sin(\frac{3\pi}{10} + \frac{\pi}{5} - x)}{\sin(\frac{3\pi}{10} + \frac{\pi}{5} - x) + \cos(\frac{3\pi}{10} + \frac{\pi}{5} - x)} dx$$

[Since property $\int_a^b f(x)dx = \int_a^b f(a+b-x)dx$]

$$I = \int_{\frac{\pi}{5}}^{\frac{3\pi}{10}} \frac{\sin(\frac{\pi}{2} - x)}{\sin(\frac{\pi}{2} - x) + \cos(\frac{\pi}{2} - x)} dx$$

$$I = \int_{\frac{\pi}{5}}^{\frac{3\pi}{10}} \frac{\cos x}{\sin x + \cos x} dx \dots\dots\dots (2)$$

Adding (1) & (2),

$$2I = \int_{\frac{\pi}{5}}^{\frac{3\pi}{10}} \frac{\sin x + \cos x}{\sin x + \cos x} dx$$

$$2I = \int_{\frac{\pi}{5}}^{\frac{3\pi}{10}} 1 dx$$

$$2I = \left[x \right]_{\frac{\pi}{5}}^{\frac{3\pi}{10}}$$

$$\therefore I = \frac{\pi}{20}$$

- Q. 40 The probability of hitting a target by A and B are $\frac{1}{3}$ and $\frac{2}{5}$ respectively. A shoots at the target once, followed by B shooting at the target once. The probability of hitting the target is
- (A) $\frac{2}{15}$ (B) $\frac{5}{15}$ (C) $\frac{8}{15}$ (D) $\frac{9}{15}$

Answer (D)

Solution .

$$\text{Let, } p(A) = \frac{1}{3} \text{ \& } p(B) = \frac{2}{5}$$

$\therefore$ The probability of hitting the target is:-

$$P = p(A) p(B) + p(A) p(\bar{B}) + p(\bar{A}) p(B)$$

$$P = \frac{1}{3} \times \frac{2}{5} + \frac{1}{3} \times \left(1 - \frac{2}{5}\right) + \left(1 - \frac{1}{3}\right) \times \frac{2}{5}$$

$$\therefore P = \frac{9}{15}$$

- Q.41 The value of k for which the points (5,5), (k,1), (10,7) lie on a straight line is
- (A) -5 (B) +5 (C) -2 (D) +2

Answer (A)

Solution .

Equation of line is given by :-

$$y = mx + c$$

Where, m - slope of the line

c - the y -intercept of line

Now , for point (5, 5)

$$5 = 5m + c \dots\dots\dots(1)$$

For point (k, 1)

$$1 = km + c \dots\dots\dots(2)$$

For point (10, 7)

$$7 = 10m + c \dots\dots\dots(3)$$

Solving (1) & (2),

$$m = \frac{2}{5}, c = 3$$

from equation (2)

$$1 = \frac{2}{5} \times k + 3$$

$$\therefore k = -5$$

Q.42 A steel wire ropes of 25 mm diameter weighing 37 N/m has 6 strands of 7 wires each. The diameter and tensile strength of each wire are 2.5 mm and 1800 MPa, respectively. The factor of safety for raising a cage of weight 60 kN from a depth of 200 m is

- (A) 5.60 (B) 4.50 (C) 4.25 (D) 4.15

Answer (A)

Solution .

We have,

$$\text{Factor of safety} = \frac{\text{Strength of rope}}{\text{Weight on rope}}$$

Now, strength of rope = $(6 \times 7) \times \text{strength of wire} \times \text{cross sectional area of wire}$

$$\text{Strength of rope} = (6 \times 7) \times 1800 \times 10^6 \times 3.14 \times \frac{(2.5 \times 10^{-3})^2}{4}$$

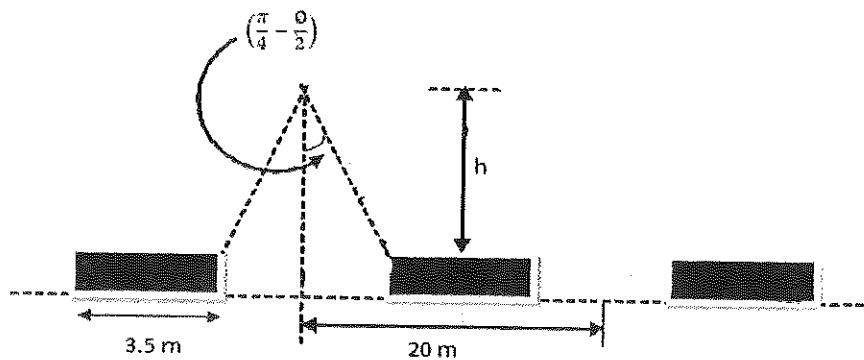
$$\therefore \text{Strength of rope} = 370912.5 \text{ N}$$

$$\text{Weight on rope} = 60 \times 10^3 + 37 \times 200$$

$$\therefore \text{Weight on rope} = 67400 \text{ N}$$

$$\therefore \text{Factor of safety} = \frac{370912.5}{67400} = 5.5$$

Q.43 In block caving operation the draw points are placed at 20 m center to center, with the pillar width 3.5 m as shown in the figure below. The muck is assumed to have zero cohesion and 35° angle. The height of draw cone (h) in m is



(A) 12.5

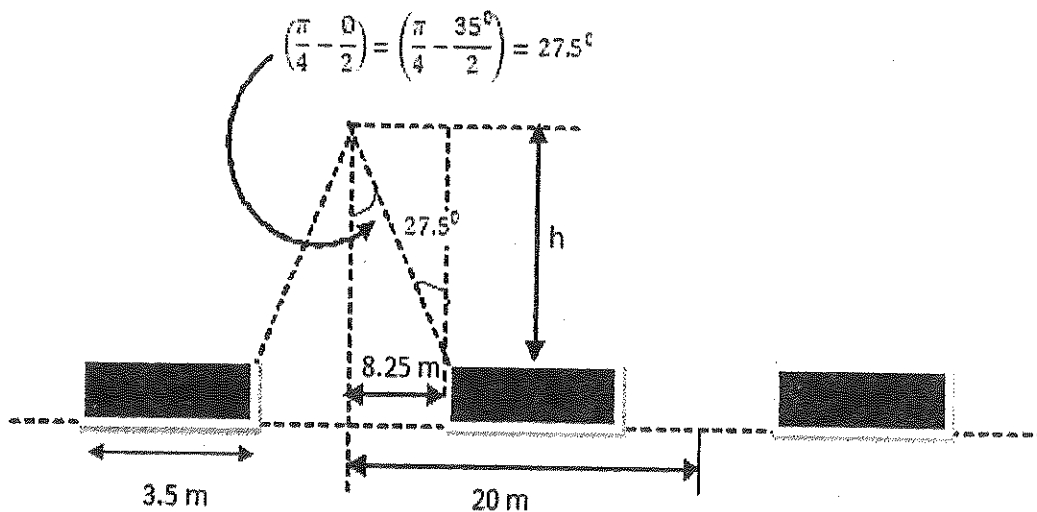
(B) 14.6

(C) 15.8

(D) 16.5

Answer (C)

Solution .



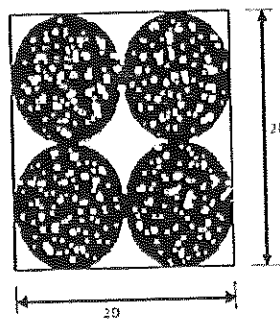
In Δabc ,

$$\tan 27.5^\circ = \frac{8.25}{h}$$

$$\therefore h = 15.8$$

Common data for questions 44 and 45 :

The granular media in ore bin is assumed to be of regular spherical shape represented by the geometry as shown in the figure. The unit weight of solids is 25 kN/m^3 .



Q. 44 The void ratio is

(A) 0.91

(B) 0.84

(C) 0.78

(D) 0.69

Answer (A)

Solution .

$$\text{Void ratio} = \frac{\text{volume of void}}{\text{volume of solid}}$$

$$\text{Volume of solid} = 8 \times \frac{4}{3} \times \pi \times \frac{D^3}{8} = \frac{4}{3} \times \pi \times D^3$$

$$\text{Volume of void} = 8D^3 - 8 \times \frac{4}{3} \times \pi \times \frac{D^3}{8}$$

$$= 8D^3 - \frac{4}{3} \times \pi \times D^3$$

$$\therefore \text{Void ratio} = \frac{8D^3 - \frac{4}{3} \times \pi \times D^3}{\frac{4}{3} \times \pi \times D^3}$$

$$\therefore \text{Void ratio} = 0.91$$

Q.45 The dry density in kN/m^3 is

(A) 13.09

(B) 12.50

(C) 11.74

(D) 10.87

Answer (A)

Solution .

$$\text{Dry density} = \frac{\text{weight}}{\text{volume}}$$

$$\text{And unit weight} = \frac{\text{weight}}{\text{volume}}$$

$$\text{Or weight} = \text{unit weight} \times \text{volume}$$

$$= 25 \times 8 \times \frac{4}{3} \times \pi \times \frac{D^3}{8}$$

$$= 104.67 D^3$$

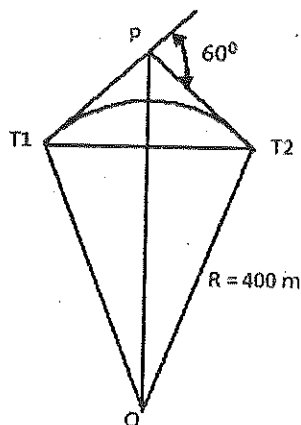
$$\text{And volume} = 8D^3$$

$$\text{Dry density} = \frac{\text{weight}}{\text{volume}} = \frac{104.67 D^3}{8D^3}$$

$$\text{Dry density} = 13.09 \text{ kN/m}^3$$

Common data for question 46 and 47

Match the elements of a simple curve as given in the figure below.



Q. 46 The tangent length in m is

(A) 215.5

(B) 220.4

(C) 228.4

(D) 230.9

Answer (D)

Solution .

We have ,

$$\text{Tangent length (L) of simple curve} = R \tan \frac{\Delta}{2}$$

Where, R- radius of curve

Δ - intersection angle

$$L = 400 \tan \frac{60^\circ}{2}$$

$$L = 230.9 \text{ m}$$

Q.47 The length of the long chord in m is

(A) 375

(B) 400

(C) 415

(D) 450

Answer (B)

Solution .

We have ,

$$\text{Length of long chord (C)} = 2R \sin \frac{\Delta}{2}$$

$$C = 2 \times 400 \times \sin \frac{60^\circ}{2}$$

$$C = 400 \text{ m}$$

Statement for linked Answer Questions 48 and 49:

A longwall panel with a face height of 3 m and face length of 150 m is worked in 3 shifts per day employing 40 men per shift. The depth of web of the shearer cutting coal is 0.5 m. The unit weight of the coal is 1.4 tonne/m^3 . Two full face cuts are executed per shift.

Q.48 The daily production from the panel in tonnes is

(A) 945

(B) 1240

(C) 1890

(D) 2530

Answer (C)

Solution .

Daily production of shearer in tonnes = (number of shift) $\times$ (two full face cuts) $\times$ (face length) $\times$ (face height) $\times$ (web depth) $\times$ (unit weight of coal)

$$\begin{aligned}\text{Daily production of shearer in tones} &= 3 \times 2 \times 150 \times 3 \times 0.5 \times 1.4 \text{ tonnes} \\ &= 1890 \text{ tonnes}\end{aligned}$$

Q.49. The panel OMS in tonnes is

- (A) 12.75 (B) 15.75 (C) 8.75 (D) 5.25

Answer (B)

Solution .

$$\text{Panel OMS in tonnes} = \frac{\text{production in three shift}}{\text{employing in three shift}}$$

$$\text{Panel OMS in tonnes} = \frac{1890}{40 \times 3} = 15.75 \text{ tonnes}$$

Statement for linked Answer Questions 50 and 51:

Air at a density of 1.2 kg/m^3 flows in a straight duct such that the velocity at the centre is 12.5 m/s . The method factor for the velocity profile is known to be 0.80 .

Q .50 The velocity pressure value in the duct in Pa is

- (A) 31 (B) 47 (C) 60 (D) 83

Answer (C)

Solution .

$$\text{Velocity pressure } (P_v) = \frac{1}{2} \times v^2 \times \rho$$

$$P_v = \frac{1}{2} \times (12.5 \times 0.80)^2 \times 1.2$$

Note – method factor is the ratio of true velocity to the observed velocity.

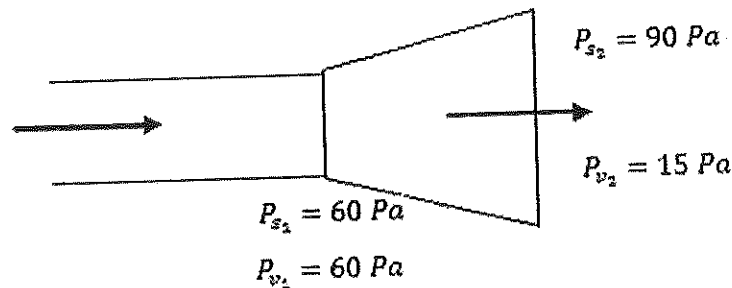
$$P_v = 60 \text{ Pa}$$

Q.51 The air flow encounters a symmetric expansion such that the cross-sectional area of the duct becomes double. The static pressure value at the inlet and outlet of the expansion are 60 Pa and 90 Pa, respectively. Neglecting friction, the shock pressure loss on account of expansion in Pa is

- (A) 15 (B) 22 (C) 38 (D) 46

Answer (A)

Solution .



$$\therefore V_2 = \frac{Q}{A_2} = \frac{Q}{2A_1} = \frac{A_1 V_1}{2A_1} = \frac{V_1}{2} = \frac{10}{2} = 5 \text{ m/sec}$$

$$\therefore P_{v_2} = \frac{1}{2} \times v_2^2 \times \rho = \frac{1}{2} \times 5^2 \times 1.2 = 15 \text{ Pa}$$

Now,

$$\text{Total Inlet pressure} = 60 + 60 = 120 \text{ Pa}$$

$$\text{Total outlet pressure} = 15 + 90 = 105 \text{ Pa}$$

Therefore,

$$\text{Shock pressure loss} = 120 - 105 = 15 \text{ Pa}$$

Q.52 Which of the following options is the closest in meaning to the word below:

Exhert

(A) urge

- (B) condemn
- (C) restrain
- (D) scold

Answer (A)

Q.53 The questions below consists of a pair of related words followed by four pairs of words.

Select the pair that best expresses the relation in the original pair.

Preamble : constitution

- (A) amendment : law
- (B) prologue : play
- (C) episode : serial
- (D) plot : story

Answer (B)

Q. 54 Choose the most appropriate word from the options given below to complete the following sentence :

The committee wrote a _____ report, extolling only the strength of the proposal.

- (A) reasonable
- (B) supportive
- (C) biased
- (D) fragmented

Answer (C)

Q.55 Choose the most appropriate word from the options given below to complete the following sentence :

If the country has to achieve real prosperity, it is _____ that the fruits of the progress reach all, and in equal measure.

- (A) inevitable
- (B) contingent

- (C) oblivious
(D) imperative

Answer (D)

- Q.56 A person invests Rs. 1000 at 10% annual compound interest for 2 years. At the end of two years the whole amount is invested at an annual simple interest of 12% for 5 years. The total value of the investment finally is :
- (A) 1776 (B) 1760 (C) 1920 (D) 1936

Answer (D)

Solution .

We have,

$$\begin{aligned}\text{Compound interest} &= P \left[1 + \frac{R}{100} \right]^T \\ &= 1000 \left[1 + \frac{10}{100} \right]^2 \\ &= \text{Rs. } 1210\end{aligned}$$

$$\begin{aligned}\text{And simple interest} &= \frac{P.R.T}{100} \\ &= \frac{1210 \times 12 \times 5}{100} \\ &= \text{Rs. } 726\end{aligned}$$

$$\begin{aligned}\therefore \text{Total investment} &= 1210 + 726 \\ &= 1936 \text{ rupees}\end{aligned}$$

- Q.57 The ban on smoking in designated public places can save a large number of people from the well know effects of environmental tobacco smoke. Passive smoking seriously impairs respiratory health. The ban rightly seeks to protect non- smokers from its all effects. Which of the following statements best sums up the meaning of the above passage.
- (A) Effects of environmental tobacco are well known

- (B) The ban on smoking in public places protects the non smokers
 (C) Passive smoking is bad for health
 (D) The ban on smoking in public places excludes passive smoking

Answer (B)

- Q. 58 Given the sequence A,B,B,C,C,C,D,D,D,D,.....etc., that is one A, two Bs, three Cs, four Ds, five Es and so on, the 240th letter in the sequence will be :

- (A) V (B) U (C) T (D) W

Answer (A)

Solution .

We know, sum of n natural numbers is

$$S = \frac{n(n+1)}{2}$$

$$\therefore 240 = \frac{n(n+1)}{2}$$

$$\text{Or } n^2 + n - 480 = 0$$

$$\therefore n = 22$$

Therefore, 22th letter will be V.

- Q.59 Consider the set of integers {1, 2, 3,, 5000}. The number of integers that is divisible by neither 3 nor 4 is :

- (A) 1668 (B) 2084 (C) 2500 (D) 2916

Answer (C)

Solution .

When, 5000 is divided by 4 then quotient = 1250

5000 is divided by 3 then quotient = 1666.67

5000 is divided by (4×3) =12 then quotient = 416.67

Hence,

$$\begin{aligned}\text{Number of integers that is divisible by neither 3 nor 4} &= 5000 - (1250 + 1666.67 - 416.67) \\ &= 2500\end{aligned}$$